

Executive Summary

Predicting Risk of Customer Loan Default

Client: SwissCredit Bank AG

Timeline: 22 weeks

SwissCredit Bank processes over 50,000 loan applications monthly across Switzerland and the EU. Current application assessment processes require 3–5 business days, resulting in a 35% attrition rate among qualified applicants and operational costs exceeding CHF 85 per manual review. Digital-native competitors approve loans within hours, capturing significant market share. At the same time, evolving regulations under the EU AI Act, GDPR, and FINMA require enhanced transparency, fairness, and oversight for AI-driven credit decisions.

We have been tasked with developing an end-to-end machine learning pipeline to predict loan default risk, balancing accuracy, interpretability, and compliance. Deliverables include a bias-aware risk prediction system, explainable model outputs, and audit-ready documentation suitable for high-risk financial AI applications.

Business Challenge

- **Processing Delays:** Manual reviews require 3–5 days, causing 35% drop-off among qualified applicants
- **Competitive Pressure:** Digital lenders approve loans in under 2 hours, increasing customer churn
- **Regulatory Exposure:** Non-compliance with EU AI Act and FINMA guidelines leads to fines of several million CHF for peer institutions
- **Operational Costs:** Manual assessment costs ~CHF 85 per application.

Objectives

1. **Reduce Decision Time:** From 3-5 days to under 30 minutes for 80% of applications
2. **Maintain Risk Standards:** Keep default rates below current 20% threshold
3. **Ensure Compliance:** Full adherence to EU AI Act, Swiss FINMA guidelines, and GDPR
4. **Improve Customer Experience:** Provide transparent explanations for all decisions

Data Assets:

- ~10 years of historical loan data (1.2M applications). Each record includes:
 - Eventual repayment outcome
 - Quarterly updates from credit bureaus
 - External demographic and economic indicators (ethnicity, education, unemployment)

Regulatory & Compliance Requirements

EU AI Act

Credit-worthiness evaluation of natural persons is designated as a high-risk AI use case. Our system must meet extra requirements: careful risk management, disciplined data governance, clear documentation, human oversight, accuracy/robustness, and post-market monitoring (with registration where applicable).

- **Design & Data:** High-risk systems need a risk-management process and data governance so inputs are appropriate for the task and quality-checked (bias, representativeness, errors)
- **Model development & testing:** You must create technical documentation describing purpose, intended use, limits, known risks, and performance; you must test for accuracy and robustness before release.
- **Deployment & operation:** There must be human oversight with clear ability to step in and overrule; you must run post-market monitoring and handle serious incident reporting via the EU database for high-risk AI

GDPR (EU Data Protection):

If you make fully automated accept/decline decisions (or similar significant effects), people have the right not to be subject to that unless specific conditions apply; in those cases, they must get meaningful information, the right to human review, and a way to contest.

- **Automated Decision-Making Safeguards:** where automated decisions produce legal or similarly significant effects, data subjects have safeguards. These include the right to obtain human intervention, express their point of view, and contest the decision; people should receive meaningful information about the logic involved.
- **Special-Category Data & Privacy Controls:** special-category data (e.g., ethnicity) is restricted and proxies must be handled carefully.

FINMA (Swiss Requirements) & revFADP:

Swiss supervisors expect clear accountability, understandable outputs, controlled model changes, and ongoing monitoring for AI in finance. Further, when a decision is made automatically (no human in the loop) that significantly affects someone, you must inform them and give them a way to have a person review it.

- **Explainability:** Outputs should be understandable to decision-makers; state limits and uncertainties to avoid over-reliance.
- **Lifecycle & third parties:** Changes (re-training, new data) must be documented and reviewed; if you use external tools or services, you must assess and monitor them.
- **Right to review:** On request, the controller must have a natural person review the decision

- **Data Protection & Non-Discrimination:** revFADP requires lawful, proportionate processing; Swiss financial supervision expects transparent, non-discriminatory outcomes, and proof that a system adheres to these properties must be readily available, e.g., actions taken to ensure equitable outcomes

SwissCredit Bank AG Customer Transparency Policy

- SwissCredit Bank AG advertises transparency in its loan decisions. It is company policy to provide customers with clear reasons for loan decisions, including when automated tools are used for making those decisions

Stakeholders and Their Priorities

Stakeholder	Primary Concerns	Success Metrics
CEO/Board	Regulatory compliance, competitive position	Market share, compliance status
Chief Risk Officer	Model accuracy, portfolio quality	Default rates, loss provisions (Money set aside to cover expected loan losses)
Head of Retail Banking	Customer satisfaction, processing speed	Time-to-decision
Compliance Officer	Regulatory adherence, audit readiness	Zero violations, clean audits
IT Director	System reliability, integration complexity, software updates in general	Uptime, API response times
Loan Officers	Tool usability, customer explanations	Adoption rate, frequency of need for intervention
Customers	Fair treatment, quick decisions, interpretability	Approval rates by demographic, turnaround time

Project Implementation

Solution Components

Through this project, you will gain experience in the steps needed to implement a successful AI-backed credit risk assessment system. Concretely, you will implement the following pipeline components:

Credit Risk Assessment Pipeline

- Range messy financial data into machine learning-ready inputs, addressing real-world data quality concerns such as missing values, outliers, and inconsistent data types
- Engineer meaningful features that can be used for model prediction, including financial ratios (debt-to-income, payment-to-income, credit utilization) and behavioral indicators that loan officers actually use
- Implement and compare multiple classification approaches, from interpretable baselines like logistic regression to sophisticated ensemble methods
- Build evaluation frameworks using both statistical metrics (AUC-ROC, precision-recall) and business metrics (expected revenue per loan, portfolio default rate)

Bias Detection and Mitigation System

- Explore how machine learning models can perpetuate or amplify societal biases in lending decisions
- Calculate and interpret fairness metrics, including demographic parity and equalized odds
- Implement practical bias mitigation strategies such as threshold optimization and fairness-aware training objectives
- Perform model selection, weighing the trade-offs between model accuracy and fairness that are often faced by financial institutions

Model Interpretability Framework

- Generate global explanations using techniques like SHAP values, LIME, and coefficient analysis to understand what drives default predictions
- Create local explanations for individual loan decisions, essential for regulatory compliance and customer communication
- Practice case study evaluation on edge cases and unusual applicant profiles

Deployment-Ready System

- Design robust data pipelines that handle the full journey from raw application data to risk score
- Build an API that can handle new inputs in an easy-to-use tool
- Monitor model performance in the face of new data and assess drift

Project-Specific Grading Details (Deliverables)

Technical Requirements

Pipeline Assessment Criteria (What instructors will verify):

Your main deliverable is a Credit Risk Assessment Pipeline, which must demonstrate the following capabilities when tested with new loan application data:

- **End-to-end processing:** Accept raw CSV data with loan applications and output risk predictions
- **Multi-model support:** Pipeline includes at least 3 different ML algorithms with performance comparison
- **Risk management capabilities:** Adaptive classification thresholds allowing managers to adjust sensitivity based on current business priorities
- **Imbalance handling:** System demonstrates robustness to data imbalance issues, i.e., does not simply always predict the majority class
- **Fairness monitoring:** Automatic calculation and reporting of bias metrics across demographic groups
- **Interpretability output:** Generation of feature importance visualizations and individual decision explanations

Example Advanced Capabilities (For extension credit):

- Ablation studies on the effects of feature transformation/engineering choices
- Ensemble methods with documented performance improvements over baseline
- Natural language model decision explanations capabilities
- Adversarial robustness testing with synthetic edge cases

Report Requirements

Aside from the general requirements described in the Grading Rubric, for this project, it is expected that your report contain the following information:

Bias and Fairness Analysis

- **Mitigation Strategy Documentation:** Detail which bias mitigation techniques were implemented and their impact on both fairness and performance metrics
- **Performance by Demographics:** Show model metrics (accuracy, default rates) across 2 protected attributes with evidence of bias mitigation improving either demographic parity or equalized odds by X%
- **Individual Case Studies:** 3 edge cases with SHAP explanations showing why borderline applications were approved/rejected

Regulatory Compliance Assessment

- **GDPR Article 22 (Right to Explanation):** Demonstrate how your system provides meaningful information about the logic behind decisions - show one example of a decision explanation a customer would receive
- **EU AI Act Article 13 (Transparency):** Document which model interpretability method you chose and why it provides "sufficiently transparent information" for loan decisions